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Toxicity prediction and risk assessment of per- and polyfluoroalkyl substances for threatened and endangered fishes☆

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ABSTRACT

Per- and polyfluoroalkyl substances (PFASs) are severely polluted in aquatic environments and can harm aquatic organisms. Due to the limitation of conducting toxicity experiments directly on threatened and endangered (T&E) species, their toxicity data is scarce, hindering accurate risk assessments. The development of computational toxicology makes it possible to assess the risk of pollutants to T&E fishes. This study innovatively combined machine learning models, including random forest (RF), artificial neural network (ANN), and XGBoost, and the QSAR-ICE model to predict chronic developmental toxicity data of PFASs to T&E fishes. Among these, the XGBoost model exhibited superior performance, with R^2 of 0.95 and 0.81 for the training and testing sets, respectively. Internal and external validation further confirmed that the XGBoost model is robust and reliable. Subsequently, it was used to predict chronic developmental toxicity data for seven priority PFASs to T&E fishes in the Yangtze River. Acipenseridae fishes (e.g., *Acipenser dabryanus* and *Acipenser sinensis*) showed high sensitivity to PFASs, possibly due to their unique lifestyle and physiological characteristics. Based on these data, the predicted no-effect concentration (PNEC) of individual PFASs was calculated, and the risk for T&E fishes in the Yangtze River was assessed. The results indicated that the risk of PFASs to T&E fishes is low (3.85×10^{-9}~8.20×10^{-4}), with perfluorohexanoic acid (PFHxA) and perfluorooctanoic acid (PFOA) as the high-risk pollutants. The risk in the middle and lower reaches of the river is higher than in the upper reaches. This study provides a new approach for obtaining chronic toxicity data and conducting risk assessments for T&E species, advancing the protection of T&E species worldwide.

1. Introduction

As of December 2023, nearly 160,000 species have been listed in the IUCN Red List of Threatened and Endangered (T&E) Species. More than 45,300 species are threatened with extinction, with about one-quarter of these being freshwater fishes (IUCN, 2023). Several factors contribute to the extinction of T&E fishes, such as environmental pollution, water conservancy projects, overfishing, invasive alien species, and diseases. Among these, environmental pollution is considered one of the main factors leading to the extinction of T&E species (Sigmund et al., 2022). Studies have shown that road runoff during heavy rainfall in Seattle, Washington, USA contains 6PPD-quinone, a tire-derived anti-ozonant, at levels exceeding the lethal concentration 50% (LC_{50}) for *coho salmon* by 24 times, causing massive deaths of them (Tian et al., 2021). In the upper reaches of the Yangtze River, the concentration of triphenyltin (TPT) can cause skeletal deformities in 3.9% and eye abnormalities in 1.7% of juvenile *Acipenser sinensis* (Hu et al., 2009). However, due to the difficulty of directly using T&E fishes for toxicity test, the toxicity data is scarce, hindering accurate assessment of the risks posed by pollutants in aquatic environments to these species.

In recent years, the development of computational toxicology has made it possible to assess the risks of pollutants to T&E species. By applying interdisciplinary knowledge from computational chemistry, bioinformatics, and systems biology, in silico models are constructed to

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efficiently simulate environmental exposure, harm, and risks of pollutants. Traditional computational toxicology methods, such as the Quantitative Structure-Activity Relationship-InterSpecies Correlation Estimation (QSAR-ICE) model, utilize the inherent relationships between the structure and physicochemical properties of pollutants and their toxicity, as well as the toxicity correlations between species, to expand toxicity data at both pollutant and species levels. For instance, Zhang et al. combined the QSAR-ICE model with the species sensitivity distribution (SSD) method to obtain sufficient acute toxicity data and derive hazardous concentrations for 5% of species for perfluorobutanoic acid (PFBA), PFOA, perfluorobutane sulfonic acid (PFBS), perfluorohexanesulfonic acid (PFHxS), and 6:2 chlorinated polyfluorinated ether sulfonate (6:2 Cl-PFESA) (Zhang et al., 2021). However, most current studies on QSAR-ICE models have focused on acute toxicity rather than chronic toxicity, possibly due to the lack of chronic toxicity data. With the advancement of artificial intelligence, machine learning has recently gained increasing attention in the toxicology field, providing another method for obtaining toxicity data for T&E fishes. Zhou et al. used random forest (RF), support vector machine (SVM), and artificial neural network (ANN) algorithms to establish a machine learning properties-environmental conditions-multi species-toxicity prediction model (ML-PEMST), which can be used to predict the toxicity of different metal nanomaterials to various aquatic organisms such as bacteria, algae, and fish (Zhou et al., 2023). Both traditional models (RF and multitask neural networks (MNN)) and advanced Graph Convolutional Networks (GCNs) have been used to establish machine learning-based QSAR models to predict the bioactivity of PFASs (Cheng and Ng, 2019). Fan et al. used the SVM model to predict the reproductive toxicity of endocrine disruptors to *Brachydanio rerio*, *Pimephales promelas, Oryzias latipes, Daphnia magna*, and *Cerodaphnia dubia* and found that the prediction ability of the SVM model was more reliable than the ICE model (Fan et al., 2021). Both machine learning and QSAR-ICE models are reliable prediction methods for toxicity data validated by a large number of studies. However, each method has its limitations. For instance, the QSAR-ICE model may exhibit poor fitting for chronic toxicity predictions due to the scarcity of toxicity data, while machine learning methods may reduce prediction accuracy due to the imbalance in data structures and the effects of missing values in the dataset. Therefore, this study intended to compare the effectiveness of both methods in predicting chronic toxicity data for T&E fishes, in order to obtain more comprehensive and accurate prediction results.

PFASs are compounds in which one or more hydrogen atoms on a carbon chain are partially or completely replaced by fluorine atoms, possessing strong thermal stability, good chemical stability, and excellent surfactant properties. They are widely used in packaging materials, firefighting foams, metal plating, pesticides, stain and water repellents, non-stick products, electronics manufacturing, and oil extraction industries (Wang et al., 2021). Currently, PFASs are frequently detected in various environmental media and biological samples, with aquatic environments being major sinks. Between 1980 and 2002, 42,000 tons of perfluorooctane sulphonate (PFOS) were discharged into aquatic environments, about 179 times more than that released into the air (Paul et al., 2009). PFOS pollution from fluorinated effluent discharge in the Nanmen River in Taiwan, China (6.05×10^3 ng/L) is the most severe worldwide in the past 20 years (Lin et al., 2010). PFOA pollution in Xiaoqing River in Shandong, China (320,000 ng/L) is the most severe (Joerss et al., 2020), two orders of magnitude higher than the next most polluted waters, the Po River and the Brenta River in Italy (348–2284 ng/L) (Valsecchi et al., 2015). The middle and lower reaches and the estuarine delta of the Yangtze River are economically developed and densely populated areas in China, with numerous fluorochemical manufacturing plants and industries, such as electronics, machinery, chemicals, metallurgy, textiles, and food, leading to severe PFAS pollution in the water. For example, the Taihu Basin, part of the lower Yangtze River system, has higher PFOA levels (57 ng/L) than the Yellow River (3.4 ng/L), Pearl River (1.9 ng/L), Hai River (8.3 ng/L), and Huai River (8.5 ng/L) basins (Wang et al., 2018; An et al., 2021). The PFBS concentration in surface water of Tangxun Lake in Wuhan reaches 8000 ng/L, two to three orders of magnitude higher than the PFBS concentration in surface water in Reno and Las Vegas, Nevada, USA (2.22–44.7 ng/L) (Shi et al., 2018; Bai and Son, 2021). Existing studies indicate that PFASs have various toxic effects on aquatic organisms, including developmental, neurotoxic, reproductive, and endocrine disruption effects. For example, exposing *Danio rerio* embryos to 1 mg/L of PFOS significantly reduced the hatching rate at 72 h post-fertilization compared to the control group, and the hatched larvae exhibited deformities. As alternatives to long-chain PFASs, PFBA and PFBS at exposure concentrations of 14 μg/L and 1 μg/L, respectively to *Danio rerio* can exhibit endocrine-disrupting toxicity by increasing vitellogenin levels and inhibiting thyroid hormone T3 levels (Zhao et al., 2016; Gebreab et al., 2020; Li et al., 2015; Satbhai et al., 2022; Chen et al., 2014; Yan et al., 2012; Wang et al., 2013; Huang, 2022). Therefore, severe PFAS pollution in surface waters is likely to threaten the survival and reproduction of T&E fishes.

Considering the impact of exposure duration and life stages, this study uses machine learning models (RF, ANN, XGBoost) and QSAR-ICE models to construct a prediction model for the chronic toxicity of PFASs to T&E fishes. The Yangtze River Basin, one of the severely PFAS-polluted areas, is the habitat of nearly 100 T&E fishes, higher than the Nile River in Africa (>20 species) and the Mississippi River in the USA (>70 species) (International Union for Conservation of Nature, 2023; Huang and Wang, 2020). Using the Yangtze River as a case study, the constructed model was further applied to predict the chronic toxicity data of seven priority PFASs identified by the US EPA (PFOA, PFOS, PFBA, PFHxA, PFHxS, perfluorononanoic acid (PFNA), and perfluorodecanoic acid (PFDA)) to T&E fishes, thereby assessing the ecological risks (Fig. 1). This study is crucial for the global risk assessment and protection of T&E fishes.

2. Materials and methods

2.1. Machine learning model

2.1.1. Dataset establishment

According to simplified molecular input line entry system (SMILES) strings, a CF compound set was first established, including compounds with at least one —CF— moiety. This set includes many compounds that would not be considered PFASs based on the current definition (C_nF_{2n+1}—R). The main goal of constructing this CF compound set is to augment data during model building (Cheng and Ng, 2019). Then, taking chronic developmental toxicity as an example, toxicity data for freshwater fishes for all substances in the CF compound set were searched in the ECOTOX database and previous literature. Relevant information, such as exposure duration (days), life stage of the test organism (embryo, juvenile, adult), and toxicity endpoints (e.g., no observed effect concentration (NOEC), lowest observed effect concentration (LOEC), and maximum allowable toxicant concentration (MATC)), was extracted to constitute a toxicity dataset. Consequently, a total of 2034 data points were collected. The collected data were further processed. If there were multiple toxicity data points for the same substance, toxicity effect, species, life stage, and exposure duration, outliers that significantly deviated from the overall data sample were removed to ensure the variation coefficient of the experimental data was ≤15%. The geometric mean of these values was taken as the modeling data, providing sufficient robustness to support the establishment of machine learning models. Ultimately, a dataset containing 628 chronic developmental toxicity data for 110 compounds with at least one —CF— moiety across 21 freshwater fishes was established (Table S3).

For the pollutants in the toxicity dataset, the ECFP molecular fingerprints of each substance were calculated using the ChemDes website (http://www.scbdd.com/fingerprints/index/; accessed 2023-11-02). The characteristics for each organism were described using parameters

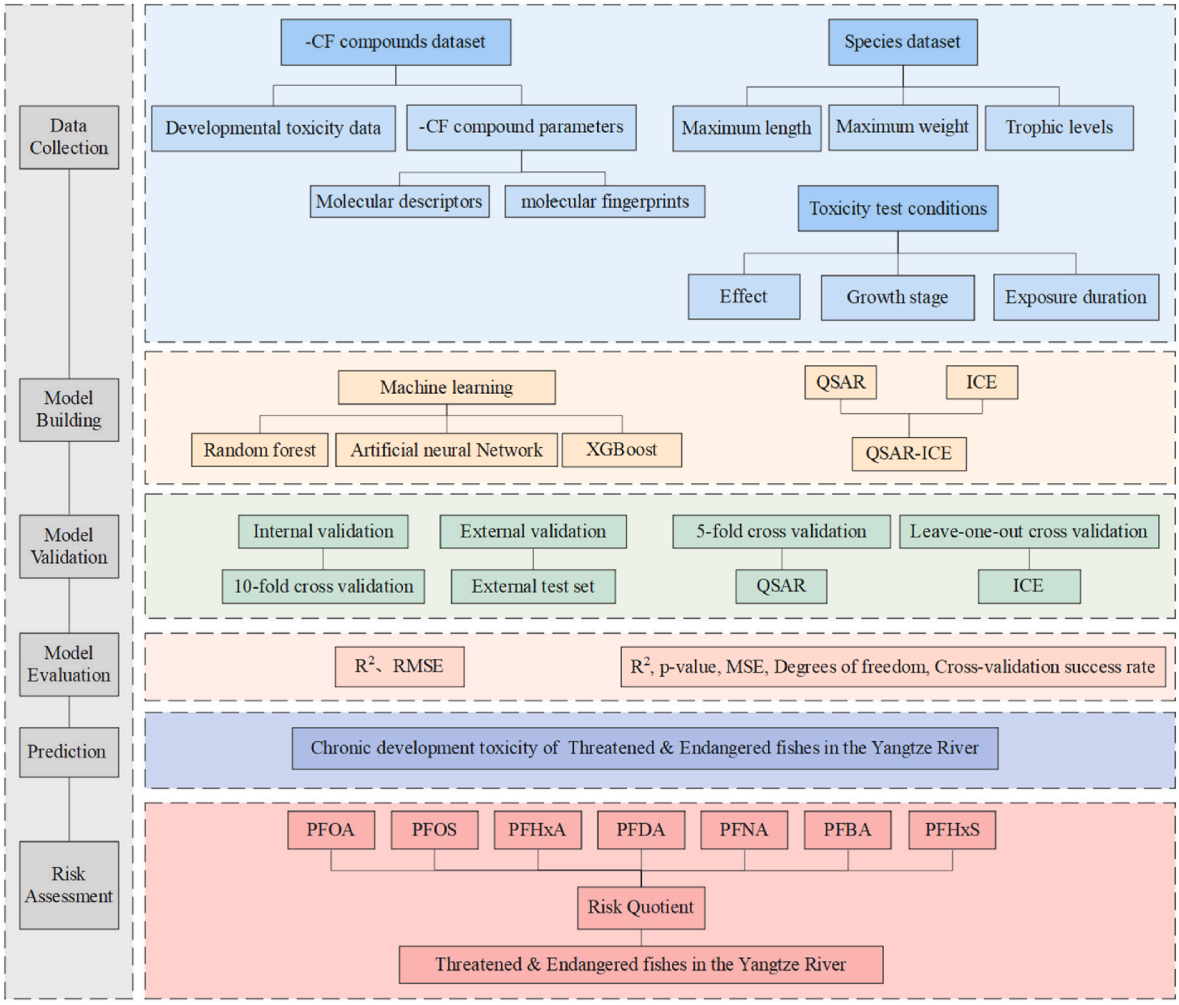

Fig. 1. Schematic illustration of the framework used in this study.

such as trophic level, life stage, weight, and length. These parameters of fishes were collected from the Fishbase website (https://www.fishbase.se/search.php; accessed 2023-12-01). Since weight data for most fishes are lacking, but length data are common, the length-weight growth equation (Bayesian length-weight) (Eq. (1)) was used to calculate the weight according to the length:

$$W = a \times L^b \tag{1}$$

where W is the fish weight in grams, L is the fish length in centimeters, and the coefficients a and b vary for different fishes, selected from the literature attached to FishBase (https://www.fishbase.se/) (Froese and Pauly, 2024).

2.1.2. Data preprocessing and model building

If a variable has more than 33% missing values, it is directly deleted. Missing values in categorical variables are assigned to a separate "missing" category. Missing values in numerical variables are replaced by the mean value of the entire variable. Numerical variables in the dataset (e.g., exposure duration, length, weight) are normalized using min-max normalization. Categorical variables (e.g., life stage, effect) are encoded using LabelEncoder. Toxicity data are Log-transformed to make it more normally distributed and mitigate the influence of outliers.

Three machine learning algorithms—RF, ANN, and XGBoost—were applied using Python 3.9 to establish prediction models for the chronic developmental toxicity of fluorine-containing compounds to freshwater fishes. The dataset was randomly divided into 80% training data for model fitting and 20% test data not involved in model building for testing the model's final performance. GridSearchCV was used to optimize the model's hyperparameters. GridSearchCV performs cross-validation on a specified parameter grid and returns the best parameter combination. The coefficient of determination (R^2) and root mean square error (RMSE) were used as the criteria for the goodness-of-fit of the model.

2.1.3. Model validation

10-fold cross-validation was used for internal validation of the three algorithm models to avoid overfitting and assess model performance. The cross-validation coefficient (Q^2_{int}) was used as an evaluation metric to check for overfitting. An external test set, not involved in the model building process, was used to evaluate the model's accuracy and robustness. The external validation coefficient (Q^2_{ext}) and the $RMSE_{ext}$ were used to assess model performance. A smaller $RMSE_{ext}$ indicates better model performance in the external validation set. The specific equations for calculating Q^2 and RMSE are as follows:

$$Q^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{2}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{m}(y_i - \hat{y})^2}{m}} \tag{3}$$

where y_i is the measured toxicity, $\hat{y}$ is the predicted toxicity, $\bar{y}$ is the mean of the measured toxicity, and m is the number of data points in the

test set.

2.1.4. Feature importance analysis

To fully understand the model and improve its interpretability, feature importance analysis is an essential step to identify which features have a greater impact on toxicity. According to section 3.1, the XGBoost model performs best in predicting chronic developmental toxicity for fluorine-containing compounds. Feature importance can be evaluated by ranking the importance coefficients. The importance coefficient is typically represented by gain, which reflects the total reduction in loss brought by each feature during splitting. Loss generally refers to some measure of error or impurity (such as Gini impurity or mean squared error). A higher gain indicates greater information gain brought by the feature during the splitting process, thereby indicating higher importance. Feature importance coefficient calculation was conducted using Python 3.9, and features were ranked based on their importance coefficients. Features with higher importance coefficients contribute more significantly to the model.

2.1.5. Model prediction

Based on model evaluation metrics in section 3.1, the XGBoost model is used to predict chronic developmental toxicity data of PFOA, PFOS, PFBA, PFHxA, PFHxS, PFNA, and PFDA to 87 species from 46 genera and 16 families of T&E fishes in the Yangtze River Basin.

2.2. QSAR-ICE model

2.2.1. Construction of ICE model

Based on the correlation of toxicity between species, an ICE model for predicting the chronic toxicity of T&E fishes in the Yangtze River was established using surrogate species. Due to the lack of toxicity studies for multiple endangered species in the Yangtze River, such as the *Acipenser dabryanus, Acipenser sinensis* and *Myxocyprinus asiaticus*, *Danio rerio* was used as the surrogate species to predict the toxicity data for five families of T&E fishes in the Yangtze River, i.e., Acipenseridae, Cobitidae, Catostomidae, Bagridae, and Cyprinidae.

First, all chronic toxicity data for *Danio rerio* and fishes from the same families as the T&E fishes in the Yangtze River were collected from the ECOTOX toxicity database (https://-CFpub.epa.gov/ecotox/) up to October 5, 2023. The endpoints of chronic toxicity effects were designated as MATC, NOEC, and LOEC, under freshwater exposure conditions with exposure duration of ≥96 h. It is important to note that due to ongoing debates on the correlation between micro-level effects (genes, cells) and macro-level toxicity effects (individual, population), toxicity data at the gene and cell levels were not used for the ICE model construction.

Next, the collected toxicity data were processed. For each substance and species, the MATC for a specific chronic toxicity effect was calculated as the geometric mean of NOEC and LOEC from the same experiment. When multiple chronic toxicity data for the same substance and species were available for the same chronic toxicity effect, outliers that significantly deviated from the overall data sample were removed, ensuring the coefficient of variation (CV) of the experimental data was ≤15%. The geometric mean of the remaining data was then taken to establish the ICE model (Table S4).

There may be multiple toxicity data points for the same species. For each species, the chronic toxicity value (CTV, μg/L), including MATC, NOEC, and LOEC, are used to calculate the species mean chronic value (SMCV, μg/L) using Eq. (4).

$$SMCV_i = \sqrt[p]{CTV_{i,1} \times CTV_{i,2} \times \ldots CTV_{i,p}} \quad (4)$$

where p is the number of CTVs (dimensionless), i is a specific species (dimensionless).

A family may contain one or more species. The SMCVs of a substance obtained for each family are geometrically averaged to obtain the family mean chronic value (FMCV, μg/L) for each family using Eq. (5).

$$FMCV_j = \sqrt[q]{SMCV_{j,1} \times SMCV_{j,2} \times \ldots SMCV_{j,q}} \quad (5)$$

Based on the processed data for modeling, the SMCV and FMCV included in the calculation were taken as common logarithms to obtain Log_{10} (SMCV) and Log_{10} (FMCV), respectively. Using Log_{10} (SMCV) for each chemical to *Danio rerio* as x-axis and Log_{10} (FMCV) for each chemical to the predicted families as y-axis, the correlation between *Danio rerio* and the predicted families was modeled in order to obtain the ICE model of chronic toxicity between *Danio rerio* and T&E fishes. According to the evaluation principles for ICE models in the US Web-ICE, the mean square error (MSE), taxonomic distance, degrees of freedom (df), R^2, and p-value were calculated to evaluate each ICE model.

The leave-one-out cross-validation was used to test the robustness and reliability of established ICE models. Specifically, one data point was removed at a time, and the remaining data were used to reconstruct the linear regression to predict the data of the removed point. If the difference between the predicted value and the original data was less than 10 times, it was called a successful cross-validation. Each point was removed in turn, and the total cross-validation success rate was ultimately calculated.

2.2.2. Construction of QSAR model

Chronic developmental toxicity data for each chemical in the CF compound set to *Danio rerio* was collected from the ECOTOX database, with endpoints including NOEC, LOEC, MATC, etc. According to OECD guidelines 212 and 215, the exposure duration was set to ≥28 days for adult fish and ≥8 days for embryos (OECD, 1998; OECD, 2000). A total of 248 toxicity data points were gathered for 21 compounds. SMILES for these compounds were obtained from the PubChem website (https://pubchem.ncbi.nlm.nih.gov/) and were then used with Chemopy, CDK, Rdkit, BlueDesc, and PaDEL modules from the Chemdes website (http://www.scbdd.com/chemdes/) to calculate the molecular descriptors of each chemical.

To simplify the QSAR model construction and avoid multicollinearity, a correlation analysis was conducted between pairs of molecular descriptors and between each descriptor and the toxicity data. Descriptors that were poorly correlated with toxicity data ($R^2 < 0.6$) and those with high inter-correlation ($R^2 > 0.9$) but lower correlation with toxicity data were excluded. A QSAR model was then developed using SPSS 27, with the selected molecular descriptors as independent variables and the logarithmic values of chronic developmental toxicity data as the dependent variable.

The model's performance was evaluated using SPSS 27, calculating metrics including the R^2, RMSE, p-value, and variance inflation factor (VIF) for each variable. 5-fold cross-validation was used to test the robustness and reliability of the established QSAR model.

2.2.3. Construction of QSAR-ICE model and toxicity prediction

The QSAR model for predicting chronic developmental toxicity of fluorine-containing compounds to *Danio rerio* was substituted into the ICE model for *Danio rerio*-T&E fishes to obtain the QSAR-ICE models for predicting chronic developmental toxicity data of the five families of T&E fishes. The value of the molecular descriptor corresponding to each PFAS was input to obtain the its chronic developmental toxicity value to the five families of T&E fishes.

2.3. Risk assessment of PFASs to T&E fishes in the Yangtze River

Predicted no-effect concentration (PNEC) is an important metric used to assess environmental risk and represents the concentration of a chemical that will not have a significant effect on the environment. In this study, the PNEC for each PFAS was obtained by dividing the most sensitive species' toxicity (Lowest Toxicity) among all T&E fishes by an assessment factor (AF), which was set as 10 (Eq. (6)) (Fan et al., 2021).

Risk quotient (RQ) is the ratio between measured effect concentration (MEC) and PNEC (Eq. (7)), which indicates the risks from individual PFASs to T&E fishes in the Yangtze River.

$$\mathrm{PNEC} = \frac{\text{Lowest Toxicity}}{\mathrm{AF}} \tag{6}$$

$$\mathrm{RQ} = \frac{\mathrm{MEC}}{\mathrm{PNEC}} \tag{7}$$

3. Results and discussion

3.1. Performance of the machine learning model

This study established a dataset including a total of 628 chronic developmental toxicity data points of 110 fluorine-containing compounds to 21 freshwater fishes. The data for *Danio rerio* (407 points) was the most accounting for 60% of the total, followed by *Pimephales promelas* (67 points) and *Oryzias latipes* (54 points) (Fig. 2). As a result, the dataset is imbalanced. During model construction, category weights were adjusted, with higher weights assigned to samples from categories with less data and lower weights to samples from categories with more data. This adjustment was made to induce the model to focus more on the characteristics of the categories with less data. Based on the factors known to strongly influence aquatic organism toxicity in previous studies, three categories of variables were integrated to predict the chronic developmental toxicity data of PFASs on T&E fishes: species information (trophic level, life stage, body length, and body weight), experimental conditions (exposure duration and toxicity effect), and chemical characteristics described by ECFP molecular fingerprints. These variables comprehensively cover the key information affecting toxicity.

Based on these, three machine learning algorithms, i.e., RF, XGBoost, and ANN, were used to build a model for predicting the chronic developmental toxicity of PFASs to T&E fishes. The results showed that the XGBoost model ($R^2 = 0.95$, RMSE = 0.40) had the best fitting effect, followed by the RF model ($R^2 = 0.89$, RMSE = 0.59), while the ANN model ($R^2 = 0.77$, RMSE = 0.83) had the worst fitting effect (Fig. 3). Internal validation of models using the 10-fold cross-validation showed that the XGBoost model ($Q^2_{int} = 0.70$) was better than RF ($Q^2_{int} = 0.67$) and ANN ($Q^2_{int} = 0.63$) and none of these models are overfitted. External validation (Fig. 3) based on the test set not involved in model construction showed that XGBoost ($Q^2_{ext} = 0.81$, $RMSE_{ext} = 0.72$) had the best prediction performance on the external test set and 85% of the differences between predicted toxicity and measured toxicity are within ±RMSE. The prediction performance of RF ($Q^2_{ext} = 0.78$, $RMSE_{ext} = 0.77$) is slightly inferior, and ANN ($Q^2_{ext} = 0.67$, $RMSE_{ext} = 0.95$) is the worst, indicating that XGBoost has strong robustness and generalization ability. In summary, the XGBoost-based prediction model is robust and reliable and was selected as the optimal model for the prediction of chronic developmental toxicity of PFASs to T&E fishes. In addition, other studies have found that XGBoost shows superior performance in toxicity prediction, possibly due to its advantage in determining nonlinear relationships and in handling both numerical and categorical variables. For example, Lei et al. compared the prediction ability of XGBoost and the advanced Graph Attention Network (GAT) model in building a QSAR model for predicting the chronic toxicity of bisphenol A (BPA) to a variety of aquatic organisms such as *Desmodesmus subspicatus*, *Oryzias latipes*, etc. The results showed that the XGBoost model had a better performance and a smaller error than the GAT model and was more adaptable to data with small and medium sample sizes (Lei et al., 2024).

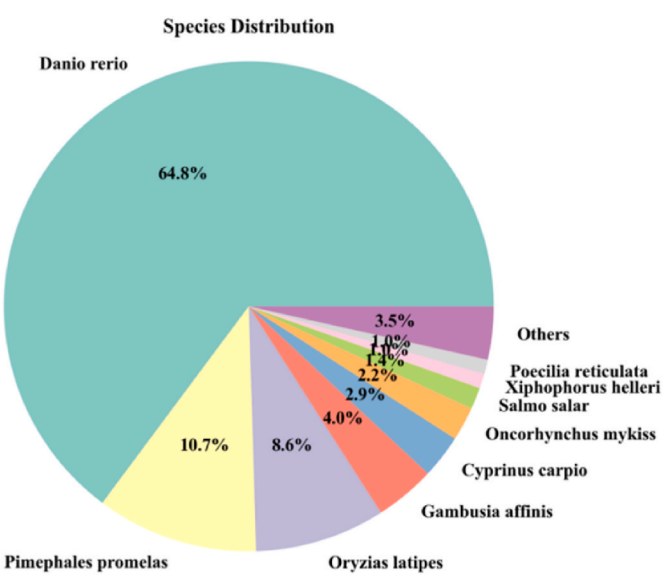

Fig. 2. Distribution of toxicity in dataset among species.

In order to fully understand the model, the XGBoost model with optimal prediction performance was analyzed for feature importance. The results showed that body weight, exposure duration, toxicity effect, and growth stage had a significant impact on the model prediction (Fig. 4). The significant influence of "body weight" on the model may be due to the toxicokinetic differences between fish of different body weights. Generally, heavier fish have slower metabolic rates and may be exposed to pollutants for longer periods of time, thus accumulating higher concentrations of pollutants. The influence of "exposure duration" on the model indicates that PFASs may accumulate more in fish with increasing exposure duration, resulting in greater effects on its physiological and biochemical processes. The influence of "toxicity effect" on the model indicates that the selection of different toxicity has a significant impact on the model prediction. In this study, the broad definition of developmental toxic effects includes growth toxicity, developmental toxicity, and morphological toxicity. Growth toxicity represents length and weight changes at any point in the life cycle. Developmental toxicity encompasses effects on tissue organization during early life stages. Morphological toxicity addresses the structure (bones) and form (organ/tissue development) changes of an organism at any stage of its life history. Each type of toxicity represents different aspects of how PFASs impact fish health. Distinguishing between the different effects is crucial for accurate prediction of the chronic toxicity. The influence of "growth stage" on the model illustrates the different sensitivities of fish to the external environment and pollutants at different growth stages. For example, juvenile stages are usually more sensitive than adult stages because they are not yet fully developed and their body systems are more susceptible to external pollutants. The growth stage determines the physiological state of the fish and its ability to resist toxicity.

3.2. Prediction of the PFAS toxicity to T&E fishes using the machine learning model

XGBoost is the best model for predicting the chronic developmental toxicity of PFASs to T&E fishes. The juvenile stage is in the early stage of fish growth and development and is commonly used to study the chronic developmental toxicity of organisms. Therefore, in this study, the XGBoost model was used to predict the chronic developmental toxicity of seven priority PFASs identified by the US EPA, namely, PFOA, PFOS, PFBA, PFHxA, PFHxS, PFNA, and PFDA, to 87 species of T&E fishes in the Yangtze River at the juvenile stage for 28 days of exposure. The results are shown in Table S1. The predicted SMCV toxicity were used to calculate the FMCV of each family according to Eq. (5), and the sensitivities of each family of T&E fishes to individual PFASs were analyzed and shown in Fig. 5.

It was found that among the seven PFASs, PFOS, PFOA, and PFHxS

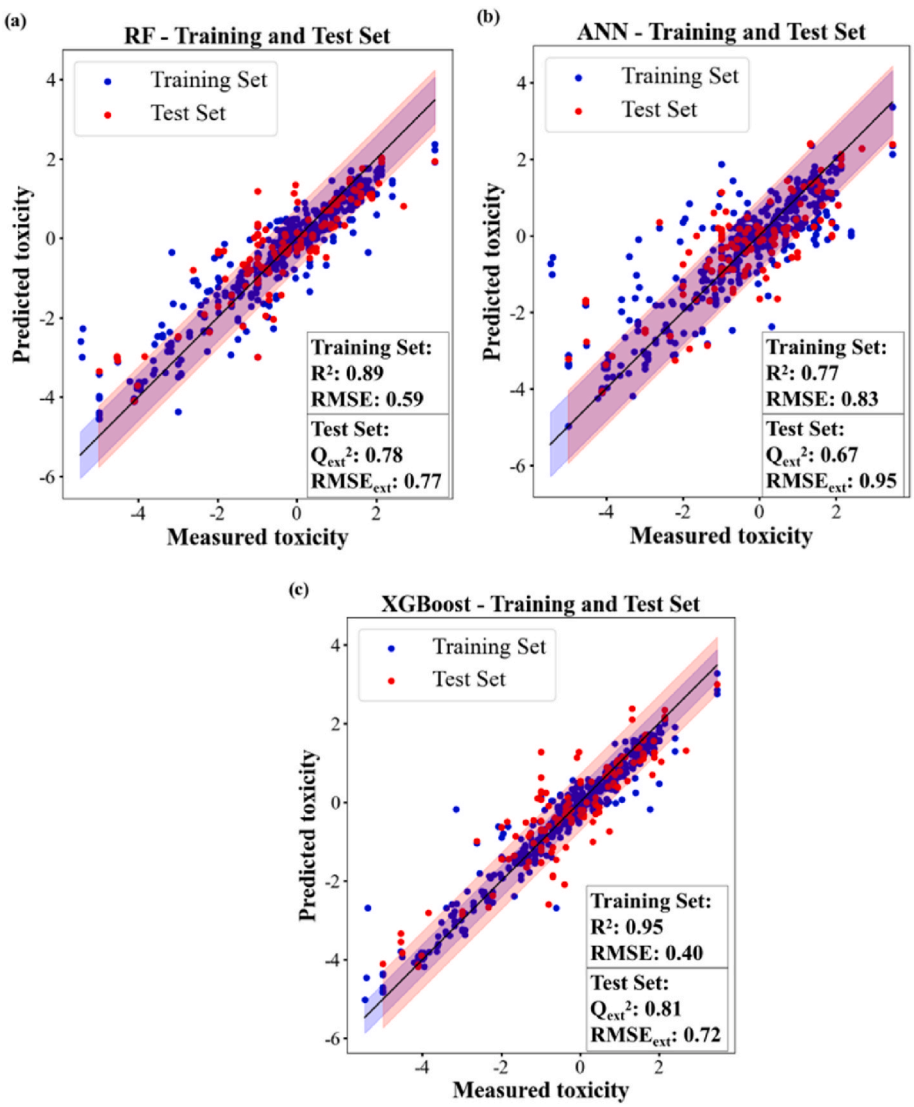

Fig. 3. Prediction performance of RF (a), ANN (b), and XGBoost (c) algorithms on training and test sets. The shaded area represents a slope of 1 with an intercept of ±RMSE.

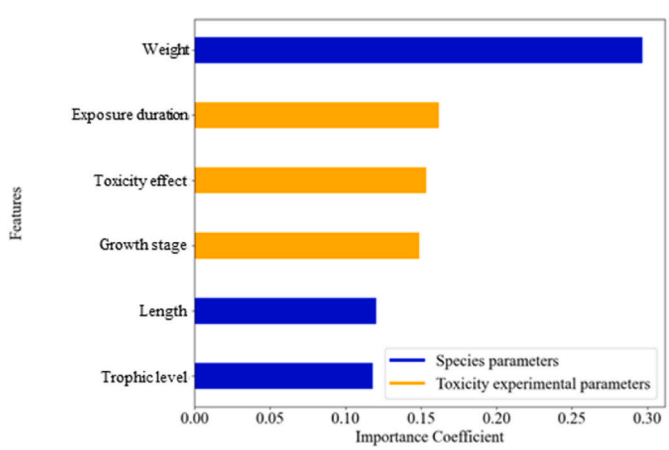

Fig. 4. Feature importance analysis of the XGBoost model.

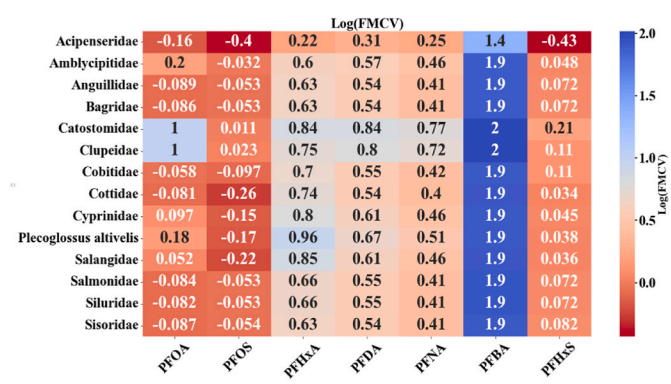

Fig. 5. Sensitivity of different families to seven PFASs (Redder color indicates lower toxicity and higher sensitivity). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

showed the highest chronic developmental toxicity to T&E fishes in the Yangtze River, PFNA, PFDA, and PFHxA showed intermediate toxicity, and PFBA showed the lowest toxicity. Acipenseridae, including *Acipenser dabryanus* and *Acipenser sinensis*, showed higher sensitivities to all seven PFASs than other fishes. Similarly, other studies have found that sturgeon are more sensitive to pollutants. For example, the study of

Xu et al. indicated that *Acipenser ruthenus* was more sensitive to BPA and Tetrabromobisphenol A than common freshwater aquatic organisms, at less than 2% and 3% on the SSD curves, respectively (Xu et al., 2022); the comparation on the sensitivity (LC_{50} 96h) between T&E species, such as the *Acipenser oxyrhynchus*, the *Acipenser brevirostrum* and *Scaphirhynchus platorynchus*, and common species (*Oncorhynchus mykiss*) showed that early life stage sturgeon was relatively sensitive to pentachlorophenol (PCP) and nonylphenol (NP), whereas the *Acipenser brevirostrum* was exceptionally sensitive to PCP (Dwyer et al., 2000; Dwyer et al., 2005). Therefore, sturgeon is more sensitive to a variety of pollutants than other species, and exposure to PFASs, BPA, and PCP may directly lead to abnormal behavior, impaired immune system, and stunted growth and development. In addition, due to their high fat content, sturgeon may increase the accumulation of pollutants such as PFASs, thereby increasing their risk (De Solla, 2016). Sturgeons are important indicator species and vulnerable taxa for the health of aquatic ecosystems, with high economic and ecological values. The long-term exposure of hazardous pollutants may not only threaten the survival of sturgeon populations, but also pose a potential threat to the entire aquatic ecosystem. In addition to Acipenseridae, Cottidae fishes are also sensitive to PFASs (Fig. 5). The high sensitivity of Acipenseridae and Cottidae fishes to PFASs may be related to their growth cycles and dietary habits. Species of these two families are benthic, long-lived, and slow-growing, and their filter-feeding behavior increases the likelihood of ingesting particles that have adsorbed pollutants from the environment, thereby leading to their high accumulation of harmful pollutants. Generally, benthic fishes tend to accumulate more pollutants and exhibit higher toxicity responses than fishes living in the upper layers of the water column due to their long-term exposure to pollutants in the sediment (Vardy et al., 2013).

3.3. Performance of the QSAR-ICE model

The ICE models of chronic toxicity between *Danio rerio* and five families of T&E fishes were established to predict chronic toxicity data of T&E fishes. All these models fitted well, with R^2 ranging from 0.71 to 0.90 (Table 1). The success rates of the leave-one-out cross-validation ranged from 67% to 88%, indicating these models were robust.

The QSAR model for predicting the chronic developmental toxicity of PFASs to *Danio rerio* is shown in Eq. (8), for which three molecular descriptors, i.e., *ATSC6v*, *MATS5s*, and *minHBa*, are used. *ATSC6v* is the van der Waals volume-weighted central Brotto-Morrow autocorrelation; *MATS5s* is the Moran autocorrelation at a lag (lag) of 5, weighted by I-states; *minHBa* is the minimum electronic state of a hydrogen-bonded acceptor. The results showed that the QSAR model was well fitted and significant for each variable ($p < 0.05$), with no multicollinearity between variables ($0 < VIF < 10$). The 5-fold cross-validation was used to test the robustness of the QSAR model. The R2 of the cross-validation was 0.80 and the MSE was 0.56, indicating that the QSAR model was reliable.

$$\text{Log10(Toxicity)} = 0.004*\text{ATSC6v}+13.574*\text{MATS5s}+0.183*\text{minHBa}-1.894 \quad (8)$$

The QSAR model for predicting chronic developmental toxicity of *Danio rerio* was substituted into the ICE models between *Danio rerio* and T&E fishes, and the QSAR-ICE models for predicting the chronic developmental toxicity data of T&E Acipenseridae, Cobitidae, Catostomidae, Bagridae, and Cyprinidae were obtained (Eqs. (9)–(13)). The performance of these QSAR-ICE models were comparable to that of other studies. For example, Hong et al. used the QSAR-ICE-SSD model to predict the toxicity of alkylphenols (APs) to aquatic organisms, with the R^2 ranging from 0.74 to 0.98 and the cross-validation success rate of >75% (Hong et al., 2022); Liang et al. used the QSAR-ICE model to predict the toxicity of linear alkylbenzene sulfonates (LAS) to aquatic organisms at the species-family level, with $R^2 \geq 0.78$, MSE ≤ 0.54, and cross-validation success rate >80% (Liang et al., 2022).

$$\text{Log(Acipenseridae toxicity)} = 0.00406*\text{ATSC6v}+13.76146*\text{MATS5s}+0.18553*\text{minHBa}-2.66161 \quad (9)$$

$$\text{Log(Cobitidae toxicity)} = 0.00355*\text{ATSC6v}+12.06308*\text{MATS5s}+0.16263*\text{minHBa}-0.99010 \quad (10)$$

$$\text{Log(Catostomidae toxicity)} = 0.00354*\text{ATSC6v}+12.00702*\text{MATS5s}+0.16187*\text{minHBa}-2.08906 \quad (11)$$

$$\text{Log(Bagridae toxicity)} = 0.00244*\text{ATSC6v}+8.27457*\text{MATS5s}+0.11155*\text{minHBa}-1.07918 \quad (12)$$

$$\text{Log(Cyprinidae toxicity)} = 0.00315*\text{ATSC6v}+10.70161*\text{MATS5s}+0.14428*\text{minHBa}-1.62794 \quad (13)$$

3.4. Prediction of the PFAS toxicity to T&E fishes using the QSAR-ICE model

Based on the QSAR-ICE model constructed in section 3.3, we predicted the chronic developmental toxicity of seven PFASs (PFOA, PFOS, PFBA, PFHxA, PFHxS, PFNA, PFDA) to five families of T&E fishes in the Yangtze River (Acipenseridae, Cobitidae, Catostomidae, Bagridae, Cyprinidae) (Table S2). The results indicated that Acipenseridae fishes were the most sensitive organisms among the five families for all seven PFASs, with toxicity values one order of magnitude lower than those of the next most sensitive family (Catostomidae). These findings are consistent with the prediction results of the XGBoost model. However, the predicted toxicity values for T&E fishes by the XGBoost model (0.18–105.79 mg/L) were about 3–50 times higher than those predicted by the QSAR-ICE model (0.05–2.31 mg/L). The discrepancies between the results of the two models may be attributed to differences in the model construction mechanisms, the used foundational toxicity data, and the uncertainty in the original data. Specifically, the QSAR-ICE model predicts toxicity based on the relationship between chemical structure and biological activity, assuming a linear relationship across species, while the XGBoost model can capture complex nonlinear patterns and adjust predictions based on diverse datasets. If the XGBoost model identifies toxicity patterns that the QSAR-ICE model does not account for, it could lead to different prediction outcomes. Moreover, the XGBoost model was built using toxicity data from 21 alternative species, while the QSAR-ICE model was based only on the correlation of *Danio rerio* toxicity with that of fishes from different families, which could introduce some errors between the two models. Additionally, chronic toxicity data vary widely due to complex mechanism and different experimental conditions. For example, in the ECOTOX database, chronic toxicity data for the same PFASs on the same freshwater

Table 1
Performance of chronic toxicity ICE models between *Danio rerio* and T&E fishes.

Families of T&E fishes	ICE model	R^2	MSE	p-value	df	Leave-one-out cross-validation success rate	Taxonomic distance
Acipenseridae	y = 1.01381x - 0.74145	0.71	0.36	3.48×10^{-2}	4	0.67	4
Cobitidae	y = 0.88869x +0.69308	0.82	0.62	1.28×10^{-2}	4	0.67	4
Catostomidae	y = 0.88456x −0.4137	0.90	0.43	4.11×10^{-3}	4	0.83	4
Bagridae	y = 0.60959x +0.07538	0.85	0.27	3.36×10^{-3}	5	0.71	4
Cyprinidae	y = 0.78839x −0.13473	0.87	0.19	1.95×10^{-11}	23	0.88	2

fishes, considering the same effect, life stage, and exposure duration, can vary by up to four orders of magnitude. These variations inevitably influence prediction results. However, compared to the differences in chronic toxicity data in the ECOTOX database, the differences between the predictions of the XGBoost and QSAR-ICE models are acceptable.

3.5. Risk assessment of T&E fishes in the Yangtze River for PFASs

Based on the chronic developmental toxicity of PFASs to 87 T&E fishes in the Yangtze River predicted by the XGBoost model (Table S1), the PNECs of seven PFASs were calculated (Eq. (7)). Consequently, the PNECs of PFOA, PFOS, PFHxA, PFDA, PFDA, PFBA, and PFHxS were 0.057, 0.023, 0.038, 0.075, 0.071, 0.922, and 0.018 mg/L, respectively. The PNECs of PFOA and PFOS were compared with the current water quality criteria of PFOA and PFOS published in the US, EU, and Australia for the protection of aquatic organisms (Fig. S1). The results indicate that the current published criteria of PFOA are 1.64–29.81 times higher than the PNEC for the protection of T&E fishes determined in this study; and the PFOS criteria published by the US EPA, as well as those from Florida and Michigan, are 1.64–6.22 times higher than the PNEC determined in this study (US Environmental Protection Agency, 2022a; US Environmental Protection Agency, 2022b; Heads of EPAs Australia and New Zealand, 2020; Boston et al., 2018; European Union, 2011; FDEP, 2020; EGLE, 2020; MPCA, 2007). This suggests that these criteria may not effectively protect T&E fishes.

Using the pollution levels of PFASs in important T&E fish habitat of the Yangtze River, including Yibin, Sichuan (YB-1∼YB-5), Yichang, Hubei (YC-1∼YC-4), Beigang, Shanghai (SH-1∼SH-5), and Poyang Lake, Jiangxi (PY-1∼PY-5), determined during 2023.5.31 to 2023.6.13, this study assessed ecological risks of the seven PFASs to T&E fishes (Fig. 6). It was found that the long-term developmental risk of ∑PFASs to T&E fishes in the Yangtze River was low (3.85×10^{-9}∼8.20×10^{-4}), of which PFHxA and PFOA had the highest risk of 1.86×10^{-4} and 1.37×10^{-4}, respectively, PFOS (3.26×10^{-5}) and PFHxS (1.27×10^{-5}) had the next highest risk, and PFDA had the lowest risk (9.69×10^{-7}). Across the four studied areas, the middle and lower reaches exhibited higher risk than the upper reaches. Specifically, the risk posed by ∑PFASs to T&E fishes followed this order: Poyang Lake (4.98×10^{-5}) > Shanghai (1.65×10^{-5}) > Yichang (2.96×10^{-6}) > Yibin (8.01×10^{-7}). The high ecological risks of PFASs in Poyang Lake can be attributed to extensive surrounding industrial activities, including fluorine chemical plants, textile manufacturing, electronics production, and food processing industries, all of which likely discharge PFAS-contaminated wastewater into nearby waters. Multiple inflowing tributaries heavily polluted with PFASs (∑PFASs: 146.2–586.2 ng/L) may also carry PFASs into the lake and exacerbate its contamination (Tan et al., 2018; Leng et al., 2021). Additionally, the lake's hydrological characteristics, such as slow water flow and prolonged retention time, may further facilitate the accumulation of PFASs. Overall, the risk of PFASs to T&E fishes in the Yangtze River was low, suggesting that the chronic developmental toxicity of PFASs to T&E fishes is not significant. However, further studies on the risks of other chronic toxicities, such as reproductive toxicity and neurotoxicity, are needed to ensure the comprehensive protection and healthy development of T&E fishes.

4. Conclusion

The lack of toxicity data limits accurate risk assessment for T&E species. In this study, the developmental toxicity of PFASs was taken as an example, and several methods were employed to predict the chronic developmental toxicity data of PFASs on T&E fishes in the Yangtze River, including the traditional QSAR-ICE model based on multiple linear regression and the machine learning models, such as RF, ANN, and XGBoost. It was found that the XGBoost model displayed the best performance and was subsequently used to predict the toxicity data of seven priority PFASs to 87 T&E fishes in the Yangtze River. The results indicated that PFOS, PFOA, and PFHxS were more toxic than other PFASs, and Acipenseridae fishes such as *Acipenser dabryanus* and *Acipenser sinensis* were more sensitive to PFASs than other fishes. The ecological risk of PFASs to T&E fishes in the Yangtze River is low (3.85×10^{-9}∼8.20×10^{-4}), but the current published water quality criteria for PFOA and PFOS may not effectively protect T&E fishes. However, it is worth noting that only the developmental toxicity was used as an example to predict the chronic toxicity data of PFASs to T&E fishes, due to the lack of data on other chronic effects, such as behavioral and reproductive toxicity. This limitation may result in an underestimation of the risk posed by PFASs to T&E fishes in the Yangtze River. More in-depth studies on the chronic toxicity effects of PFASs on aquatic organisms, such as reproductive toxicity, neurotoxicity, and endocrine disruption, are needed to support the obtain of more chronic toxicity data for T&E fishes. Overall, this study integrates both the traditional QSAR-ICE model and emerging machine learning models to obtain toxicity data for T&E species, which provided valuable data support and theoretical foundations for the risk assessment of a broader range of T&E species.

CRediT authorship contribution statement

Yuanpu Ji: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xiaolei Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Rui Wang:** Investigation, Data curation. **Jiayu Wang:** Data curation. **Xiaoli Zhao:** Writing – review & editing. **Fengchang Wu:** Writing – review & editing.

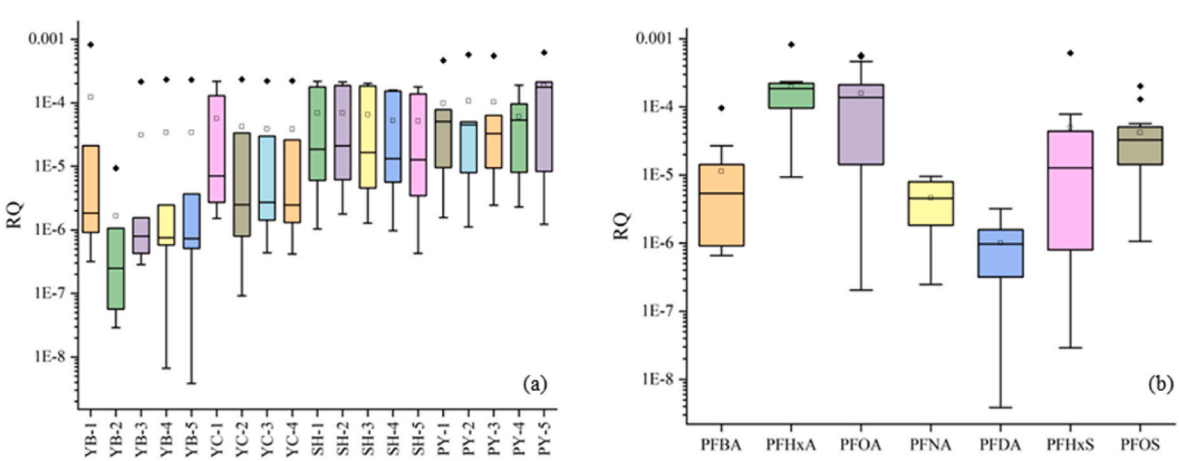

Fig. 6. RQ distribution among different areas in the Yangtze River (a) and among different PFASs (b).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2024.124920.

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